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STAR
Spatial Topography Accessibility Robot

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Texas A&M University
ESET Senior Capstone, Spring 2026

$236,451 per door

Federal civil penalty per repeat ADA violation, after the July 2025 Federal Register inflation adjustment.

8,800 federal Title III lawsuits were filed in 2024 (Seyfarth Shaw tracker), continuing a sustained 9k-11k baseline since 2018.

Manual audits cost $800-$50,000+ per facility and take days. The math does not close.

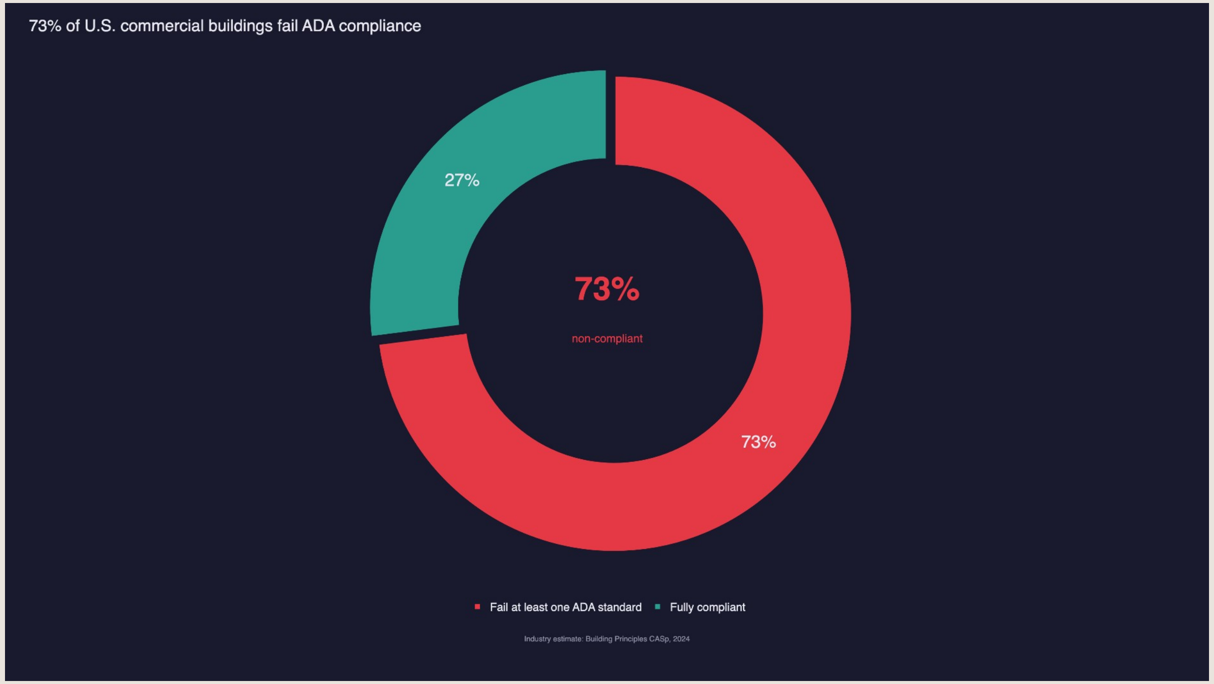
$118,225
first violation

$236,451
each subsequent

Who measures this?

Per Building Principles CASp (2024 industry estimate), 73 percent of U.S. commercial buildings fail at least one ADA standard.

12.2 percent of U.S. adults (CDC) report mobility disabilities -- approximately 31 million people.

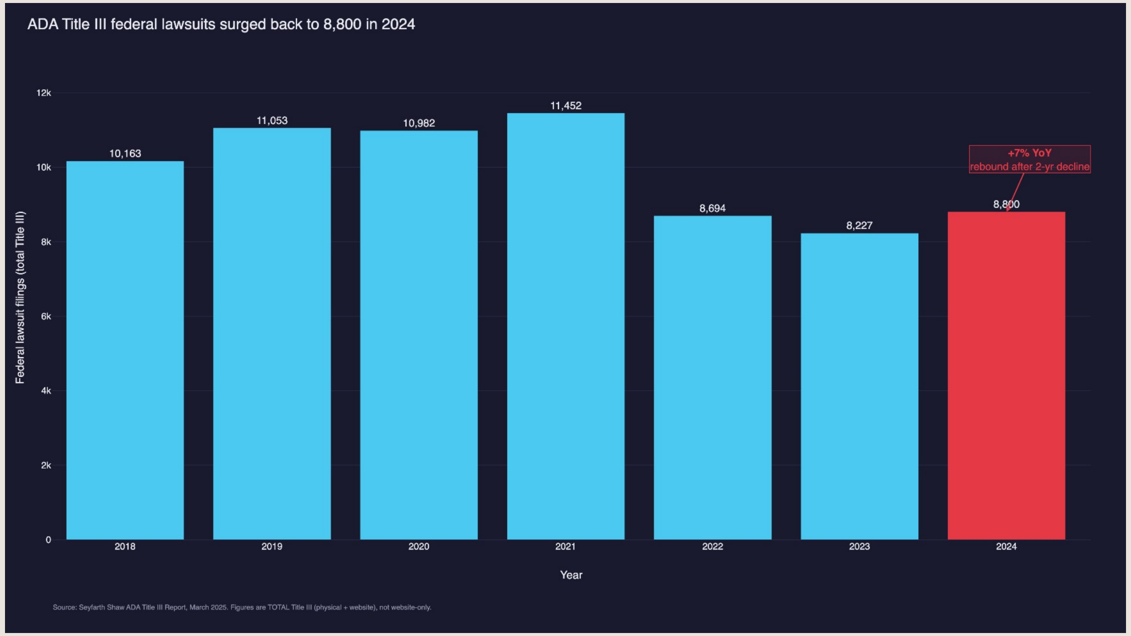


Enforcement is accelerating

Federal Title III filings: 10,163 (2018) -> peak 11,452 (2021) -> rebounded to 8,800 in 2024.

California leads with 3,252 filings in 2024; Texas filed 224.

Title II web-accessibility rule (DOJ Apr 2024, deadlines extended Apr 2026 to Apr 2027 and Apr 2028) coincided with a 7 pct year-over-year rebound.



Tech stack

HARDWARE

RPLiDAR C1 (12 m, IP54)

IMX219-83 stereo cameras

BNO055 IMU, HC-SR04 sonar

Renesas RX72N (240 MHz)

Raspberry Pi 5

DRV8263H driver x4

5S 18650 battery pack

SOFTWARE

ROS 2 Jazzy on Ubuntu 24.04

slam_toolbox (mapping mode)

Nav2 + RegulatedPurePursuit

robot_localization EKF

Go gateway with nanopb

Python compliance engine

C23 firmware on ThreadX

VALIDATION

60 firmware ctests, 100 pct libs/

74.0 pct Go gateway cov, gate 65

14 ROS 2 colcon tests

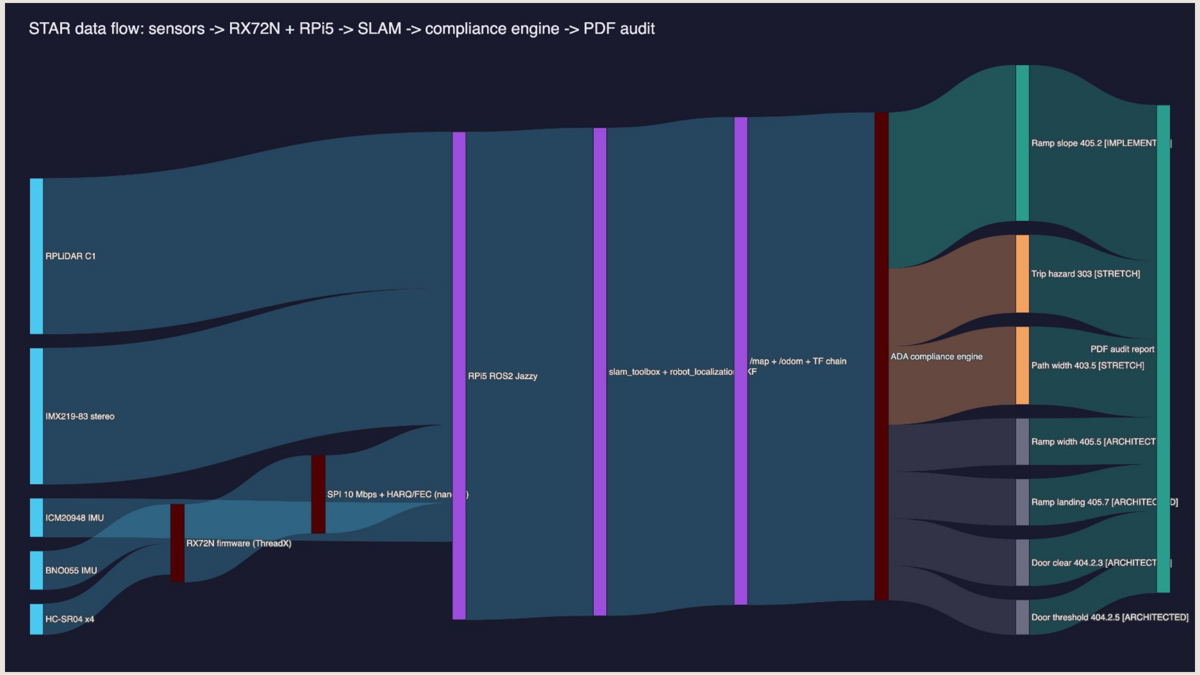
12 compliance pytest modules

9 manual PCB validation checks

Wixey WR300 +/- 0.1 deg ground truth

System architecture

- Sensors feed the RX72N for hard-real-time motor control at 250 Hz.
- RX72N talks to the Pi 5 over a framed transport: SPI 10 MHz with FEC and HARQ is the architected production link; USB-CDC ASCII at 115200 baud is the deployed link for capstone defense.
- Pi 5 runs slam_toolbox, robot_localization EKF, and Nav2. The compliance engine reads the fused map and emits a structured PDF report.



Seven ADA checks

IMPLEMENTED

Ramp slope (ADA 405.2)

RANSAC plane on LiDAR + BNO055 pitch cross-validation

Threshold 4.76 deg (1:12)

STRETCH

Trip hazard (ADA 303): vertical discontinuity > 1/4 in

Path width (ADA 403.5): cross-section along planned route

Existing but not tested due to limitations with motors:

Ramp width (405.5)

Ramp landings (405.7)

Door clear width (404.2.3)

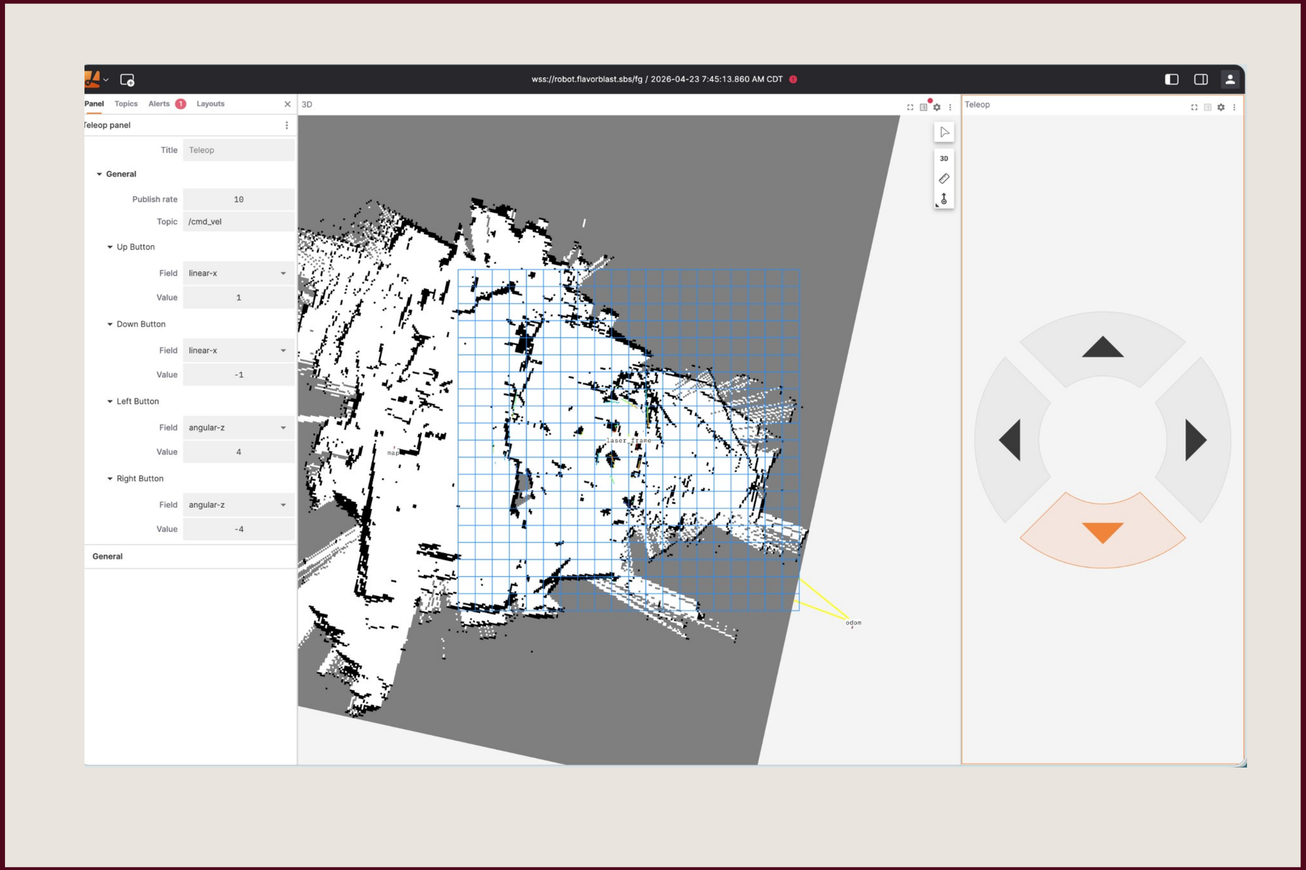
Door threshold (404.2.5)

Live demo

Robot is dispatched in MANUAL via the Grafana cockpit, then handed to AUTONOMY for frontier exploration.

On approach to a ramp, the compliance engine fits a RANSAC plane and cross-validates pitch against the BNO055.

When measured slope > 4.76 deg, the engine logs a violation flag with the ground-truth row in the PDF audit report.



Validation

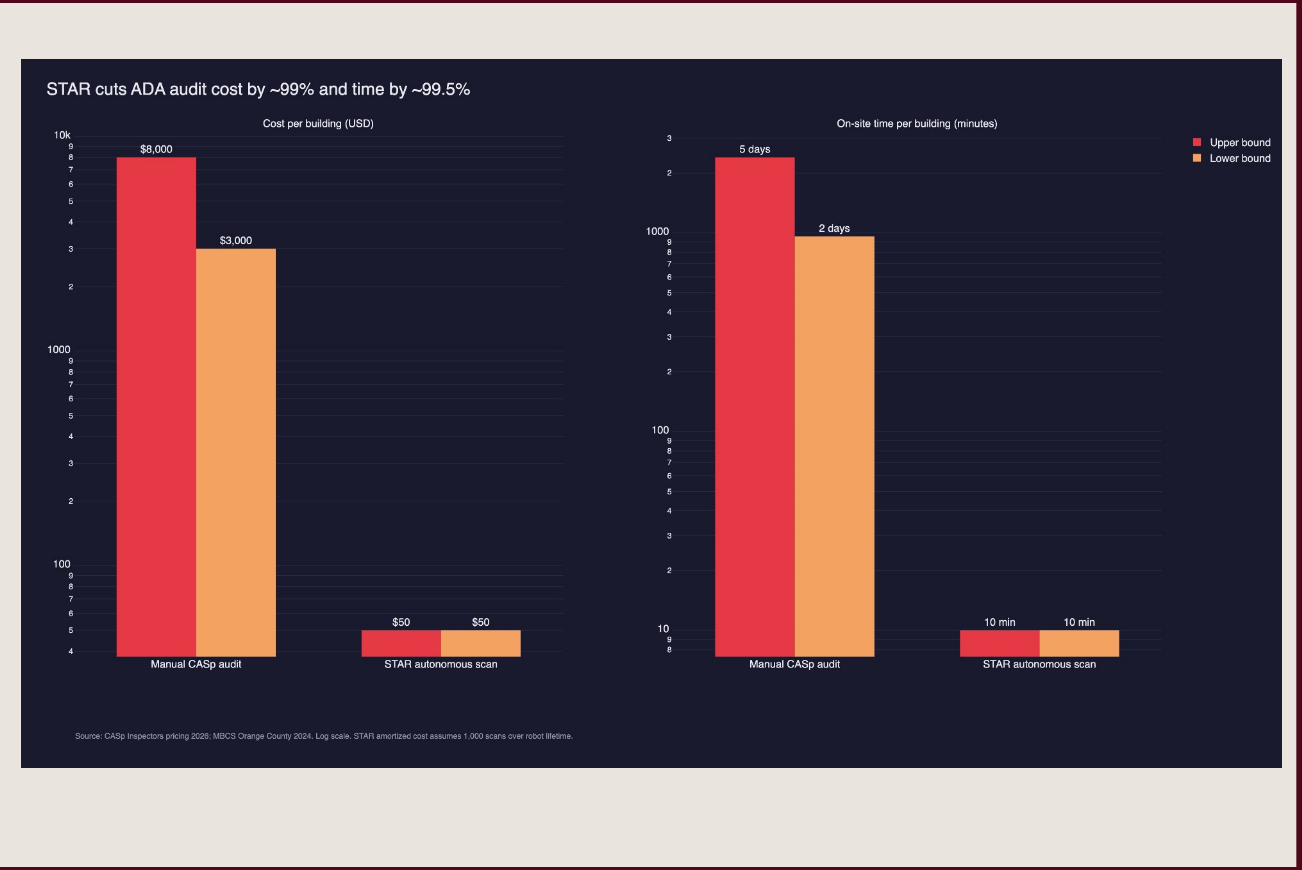
- Software: 60 firmware ctests (100 pct libs/), 74 pct Go gateway, 14 ROS 2 tests, 12 pytest modules.
- Hardware: 9 manual PCB checks all PASS: continuity, motor free-spin + loaded chassis with DC current, scoped PWM, AD2-probed comm.
- Field: autonomous frontier-exploration scan of a Thompson Hall hallway (right); demo run flags ramp-slope and path-width violations.

Cost

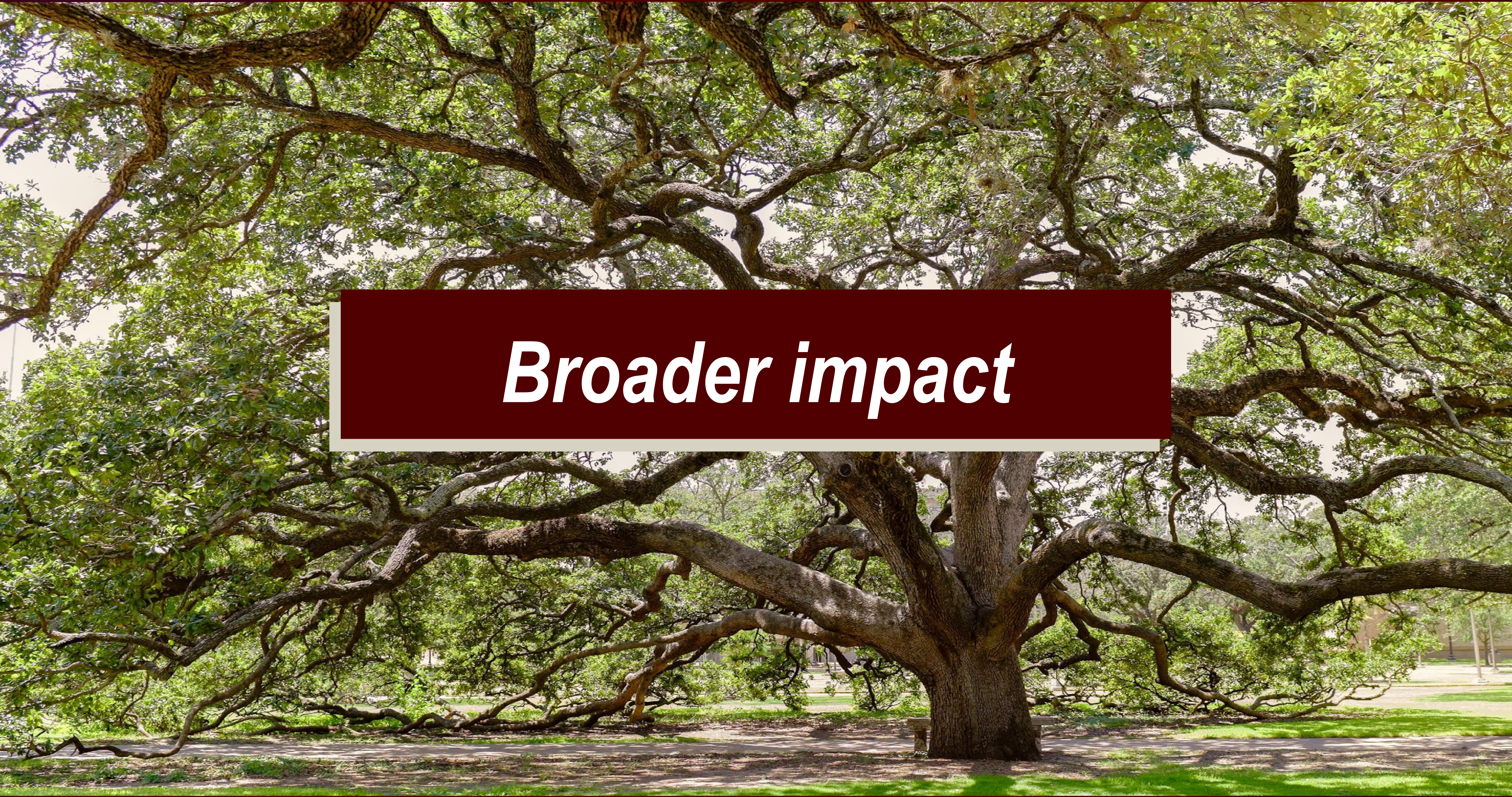
Design BOM ~$864; ordered total ~$1,249 component cost (~$1,549 with shipping). Well under the $2,000 capstone target.

Compute $182 (Pi 5, RX72N, PCB) - Sensors $270 (LiDAR, stereo, IMU, sonar) - Drivetrain $144 (motors, drivers, wheels) - Power $108 (battery, BMS, regulators) - Chassis $100

Daxbot service-as-a-product is $920k for 950 surveyed miles. STAR is sub-$2k recurring near-zero. Different economics entirely.



Broader impact



Future work

- Implement architected ADA checks: ramp width, ramp landing, door clear width, door threshold.
- Sensor fusion via RTAB-Map for richer 3D reconstruction.
- Multi-floor support with elevator detection and floor handoff.
- ML-based fixture recognition for fire extinguishers, signage, hardware.
- Outdoor GPS plus RTK for campus-scale audit campaigns.

Q&A and team

SOFTWARE

Cesar Magana, Project Manager

proto, gateway, ROS 2, UI, MATLAB

EMBEDDED AND PCB

Brighton Sikarskie

firmware, compliance engine, infra, schematic

Jeremie Hockey, PCB layout

HARDWARE AND REVIEW

Michael Norton, mechanical design

Shawn Ciaciura, schematic review

Jared Bartz, schematic review

THANK YOU

Locked Inc. - ESET 420 Senior Capstone, Spring 2026 - github.com/Locked-Inc/STAR